Algorithm for file updates in Python

Project description

This project demonstrates how Python can be used to securely maintain an allow list of IP addresses for restricted network access in a healthcare environment. The algorithm reads IP addresses from a file, converts them into a list, and checks them against a separate remove list. Any IP addresses that should no longer have access are removed. Finally, the updated list is written back to the original file, ensuring access permissions remain accurate and up to date.

Open the file that contains the allow list

```
import_file = "allow_list.txt"
with open(import_file, "r") as file:
```

Read the file contents

```
ip_addresses = file.read()
```

Convert the string into a list

```
ip_addresses = ip_addresses.split()
```

Iterate through the remove list

```
for element in remove_list:
```

Remove IP addresses that are on the remove list

```
if element in ip_addresses:
   ip_addresses.remove(element)
```

Update the file with the revised list of IP addresses

```
updated_ips = "\n".join(ip_addresses)

with open(import_file, "w") as file:
   file.write(updated_ips)
```

Summary

This algorithm begins by opening the allow list file using a `with` statement and the `open()` function, ensuring the file is handled safely. The `.read()` method converts the file contents into a string, which is then transformed into a list using `.split()`. A `for` loop iterates through a remove list and checks whether each IP address exists in the allow list. If a match is found, the `.remove()` method deletes the IP address from the list; this works correctly because there are no duplicate IP addresses. The updated list is converted back into a string using `.join()` with newline separators. Finally, the revised data is written back to the file using the `.write()` method.